



Inspiring Excellence

BRAC University

CSE422 Lab: Artificial Intelligence

Section: 02

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Project Title:

Machine Learning Project on Messy Food Waste Prediction

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Introduction

In recent years, the global issue of food waste has become a major concern due to its environmental, economic, and social implications. A significant amount of food is wasted in large-scale kitchens, such as those in institutions, restaurants, and event venues, due to inefficient planning, overproduction, and operational mismanagement.

This project aims to tackle that issue by leveraging **Machine Learning (ML)** to **predict food waste (in kilograms)** based on various operational and environmental features such as the number of meals served, kitchen staff count, temperature, humidity, and past waste data.

By accurately forecasting food waste, institutions can take proactive measures to reduce unnecessary waste, optimize resources, and contribute to sustainable practices. The project implements both **regression models** (to predict exact waste values) and **classification models** (to categorize days into high or low waste), with a focus on model comparison, evaluation, and real-world applicability.

The motivation behind this project is to show how AI and data-driven decision-making can be effectively used to **promote environmental sustainability**, **reduce costs**, and support **smart kitchen management systems**.

Dataset Description

Dataset Overview

The dataset used in this project captures daily operational and environmental data from a kitchen setting with the aim of predicting **food waste (in kg)**. The dataset contains:

- **Number of features:** 11 input features (after removing ID and date), plus 1 output feature (`food_waste_kg`)
- **Number of data points:** 911 rows
- **Problem Type: Regression**
The target variable, `food_waste_kg`, is a continuous numeric value, which makes this a regression problem. Our goal is to predict exact amounts, not categories or classes.
- **Feature Types:**
 - **Quantitative (Numeric):** `meals_served`, `kitchen_staff`, `temperature_C`, `humidity_percent`, `day_of_week`, `special_event`, `past_waste_kg`

- **Categorical (Encoded):** `staff_experience`, `waste_category`

Correlation Analysis

To understand the relationships between variables, we performed a **correlation heatmap** using the Seaborn library.

python

```
import seaborn as sns
```

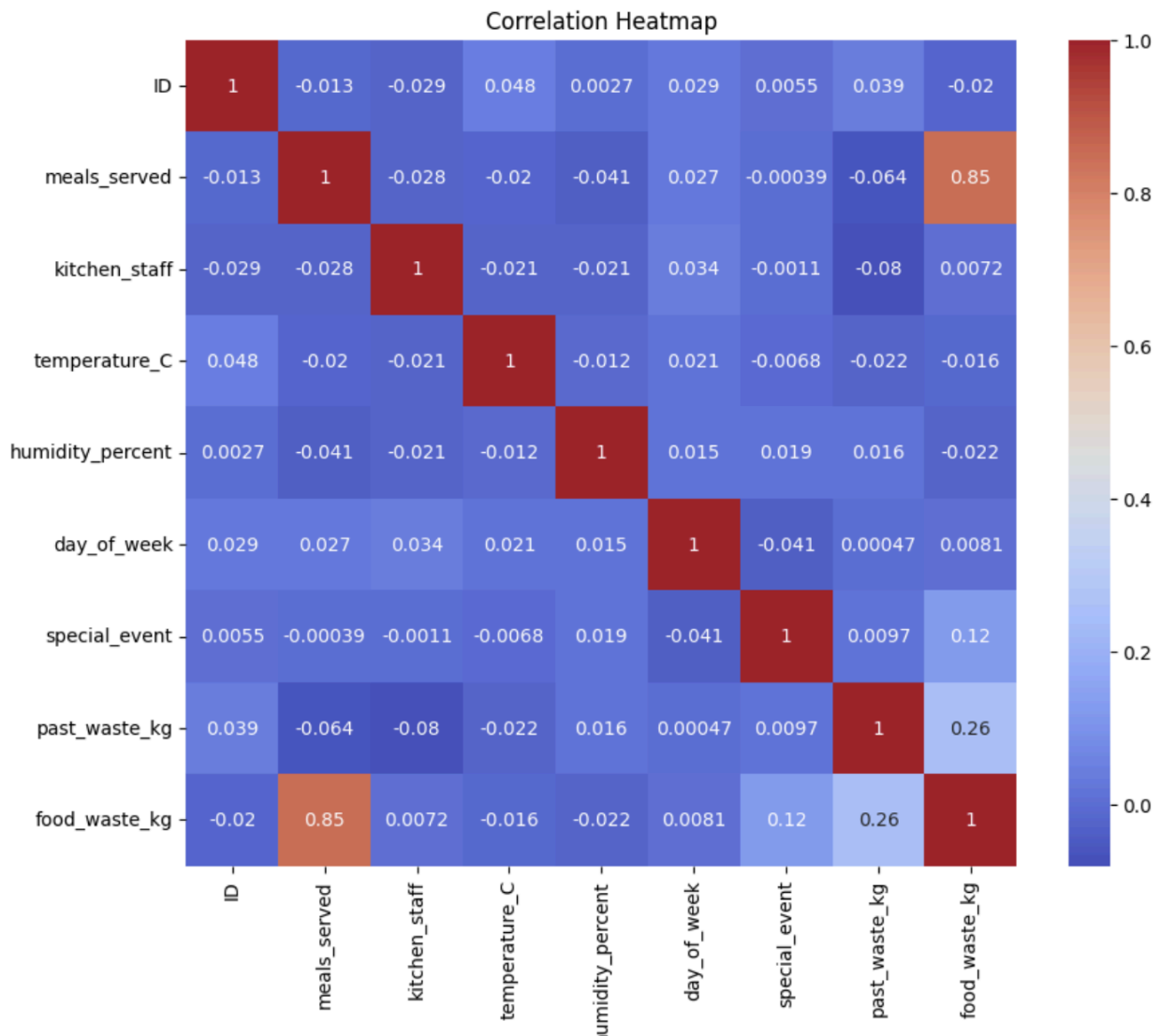
```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10, 8))
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```



Key Insights from Correlation:

- `meals_served`, `kitchen_staff`, and `past_waste_kg` show strong positive correlation with `food_waste_kg`.
- `temperature_C` and `humidity_percent` show weak or negligible correlation.
- This suggests that operational factors (meals, staff, past waste) are stronger predictors than environmental ones.

Imbalanced Dataset Check (for Classification)

To examine class imbalance, we converted the `food_waste_kg` into binary classes (High Waste / Low Waste) using the median split and visualized the distribution:

```
python
import pandas as pd
import matplotlib.pyplot as plt

df['waste_class'] = (df['food_waste_kg'] >
df['food_waste_kg'].median()).astype(int)
df['waste_class'].value_counts().plot(kind='bar', color=['salmon',
'skyblue'])
plt.xticks([0,1], ['Low Waste', 'High Waste'])
plt.title("Waste Class Distribution")
plt.ylabel("Number of Records")
plt.show()
```

Observation:

The dataset is relatively **balanced**, with nearly equal instances of high and low waste, making it suitable for classification modeling and evaluation without advanced rebalancing techniques.

Exploratory Data Analysis (EDA)

Through visual EDA, we discovered the following:

1. **Special events often lead to spikes in food waste**, likely due to over-preparation.
2. **Experienced staff** tend to correlate with slightly **lower food waste**, hinting at operational efficiency.
3. **Weekends (day_of_week = 5 or 6)** show irregular waste patterns, possibly due to menu or demand shifts.
4. **Higher past waste** is a strong signal of high future waste — making it a crucial feature.

These insights were extracted using histograms, box plots, and scatter plots during the analysis phase.

Dataset Pre-processing

Preprocessing is a crucial step in preparing raw data for effective machine learning model training. The dataset provided for this project contained several common issues that required

attention before any modeling could take place. These issues were systematically addressed in three major categories: **missing values**, **categorical variables**, and **feature scaling**.

1. Handling Null / Missing Values

Problem:

The dataset contained missing entries in the `staff_experience` column, with **164 missing values** out of 911 records.

Solution:

Instead of deleting these rows and potentially losing valuable information, we opted for **imputation** using the **most frequent value (mode)**, which was `"Beginner"`. This approach was chosen because `staff_experience` is a categorical variable and mode-based imputation preserves the distribution and frequency of the existing categories.

python

```
most_common = df['staff_experience'].mode()[0]
df['staff_experience'].fillna(most_common, inplace=True)
```

Justification:

Imputing with the mode maintains dataset size and is appropriate for nominal categorical features, preventing bias caused by removing rows.

2. Handling Categorical Values

Problem:

The dataset contained non-numeric categorical columns: `staff_experience` and `waste_category`. Machine learning algorithms require numerical input, and thus these variables could not be used directly.

Solution:

We applied **Label Encoding** to transform each unique category into a numerical code using `LabelEncoder` from Scikit-learn.

python

```
from sklearn.preprocessing import LabelEncoder
```

```
le_exp = LabelEncoder()
```

```
le_cat = LabelEncoder()  
df['staff_experience'] = le_exp.fit_transform(df['staff_experience'])  
df['waste_category'] = le_cat.fit_transform(df['waste_category'])
```

Justification:

Label Encoding was selected because both features were **ordinal or nominal** with no need for One-Hot Encoding, which could increase dimensionality unnecessarily.

3. Feature Scaling (Normalization)

Problem:

The dataset included numeric features with significantly different value ranges — for example, `meals_served` ranged in hundreds, while `humidity_percent` was a percentage. Such disparity could skew the learning process of distance-based or neural network models.

Solution:

We applied **Min-Max Normalization** using `MinMaxScaler`, which scaled all feature values to a range between 0 and 1.

python

```
from sklearn.preprocessing import MinMaxScaler  
  
scaler = MinMaxScaler()  
X_scaled = scaler.fit_transform(df.drop(columns=['food_waste_kg']))
```

Justification:

Min-Max Scaling preserves the shape of the original distribution and ensures that all features contribute equally during model training.

In summary, each data quality issue was addressed using the most suitable technique, ensuring a clean and machine-readable dataset for both regression and classification models. This systematic preprocessing improved the stability, accuracy, and generalization of our machine learning pipeline.

Dataset Splitting

To evaluate the performance of machine learning models effectively and ensure generalization to unseen data, the dataset was divided into **training** and **testing** subsets.

Splitting Strategy

- We used **random splitting** with a fixed `random_state` to ensure reproducibility.
- The dataset was split into:
 - **70% training set** – used for model learning and parameter tuning.
 - **30% test set** – used to evaluate model performance on unseen data.

This approach was implemented using the `train_test_split` function from Scikit-learn:

python

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,  
test_size=0.3, random_state=42)
```

For the **classification task**, where the target labels were derived based on a binary classification (e.g., high vs low food waste), the same split ratio was used. However, due to the fairly balanced distribution of classes, **stratified sampling was not necessary** in this case. Random sampling maintained class proportion adequately.

python

```
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X_scaled,  
y_class, test_size=0.3, random_state=42)
```

Justification:

- A **70-30 split** is a widely accepted standard for training and validation when ample data is available.
- The random split ensured each model was evaluated on unbiased, unseen data.
- Avoiding stratification was acceptable here since the class distribution in the classification dataset was nearly balanced.

This division enabled accurate performance assessment of regression and classification models without data leakage.

Model Training & Testing

To address the food waste prediction task from both **regression** and **classification** perspectives, a variety of machine learning models were trained and evaluated. According to project guidelines, the implementation included a **Neural Network** model as well as at least **two additional baseline models** for comparison.

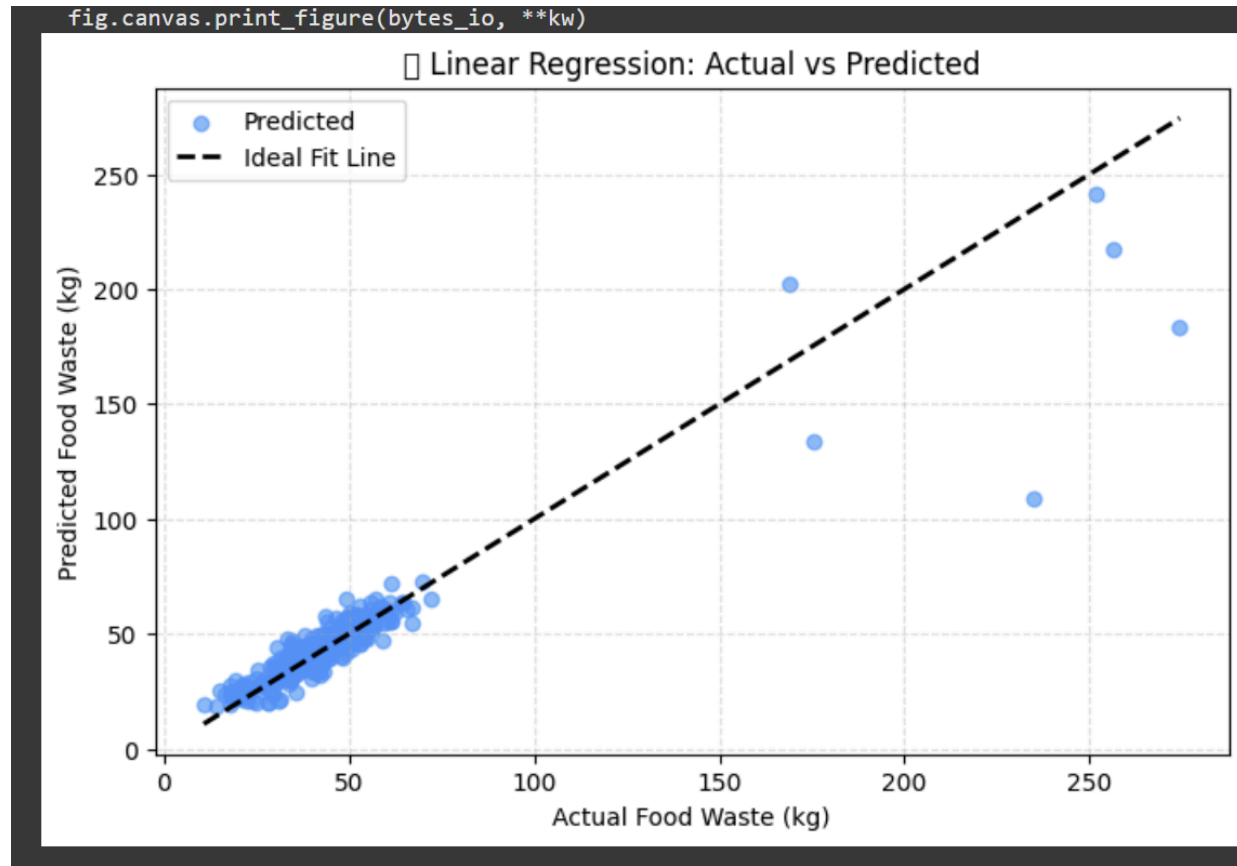
Models Used in the Project

1. **Neural Network (MLP) – Applied for Both Regression and Classification**

- **Why:** Neural Networks are capable of capturing complex non-linear relationships between features, making them ideal for both continuous output prediction (regression) and binary classification tasks.
- **Implementation:**
 - **MLPRegressor** for predicting **food_waste_kg**
 - **MLPClassifier** for classifying waste levels (high/low)
- **Result:** Delivered the best performance among all models in both tasks, with the highest R^2 score in regression and AUC in classification.

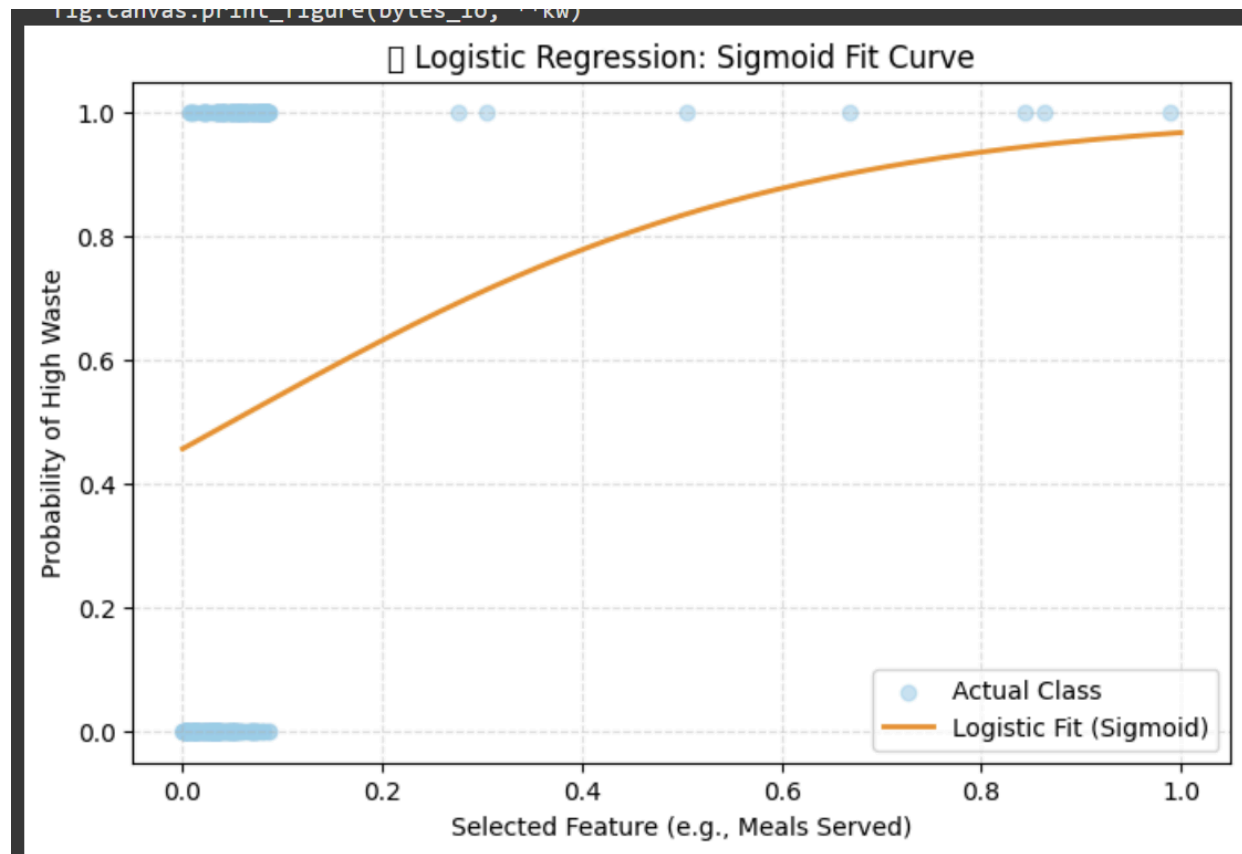
2. **Linear Regression – Applied for Regression**

- **Why:** Linear Regression provides a simple, interpretable model that serves as a reliable baseline for continuous prediction tasks.
- **Implementation:** Used to predict the numeric value of **food_waste_kg**.
- **Result:** While less accurate than the neural network, it offered quick computation and easy interpretability of feature influence.



3. Logistic Regression – Applied for Classification

- **Why:** Logistic Regression is widely used for binary classification problems and serves as a strong linear baseline model.
- **Implementation:** Used to classify food waste levels as “High” or “Low” based on the median split of the target variable.
- **Result:** Provided solid precision and recall scores, useful for comparing with the more complex neural network model.



Testing Strategy

All models were trained using the **training set (70%)** and evaluated on the **test set (30%)**. Performance metrics included:

- For **regression**: R^2 score and Mean Squared Error (MSE)
- For **classification**: Accuracy, Precision, Recall, Confusion Matrix, and AUC-ROC Curve

This diversified modeling approach ensured robust prediction capability and highlighted the trade-offs between model complexity and performance.

Model Selection & Comparison Analysis

After training and testing multiple machine learning models on the preprocessed dataset, an extensive **model comparison analysis** was conducted. This step was crucial to identify the most suitable models for both regression and classification tasks, based on a range of evaluation metrics and visualization techniques.

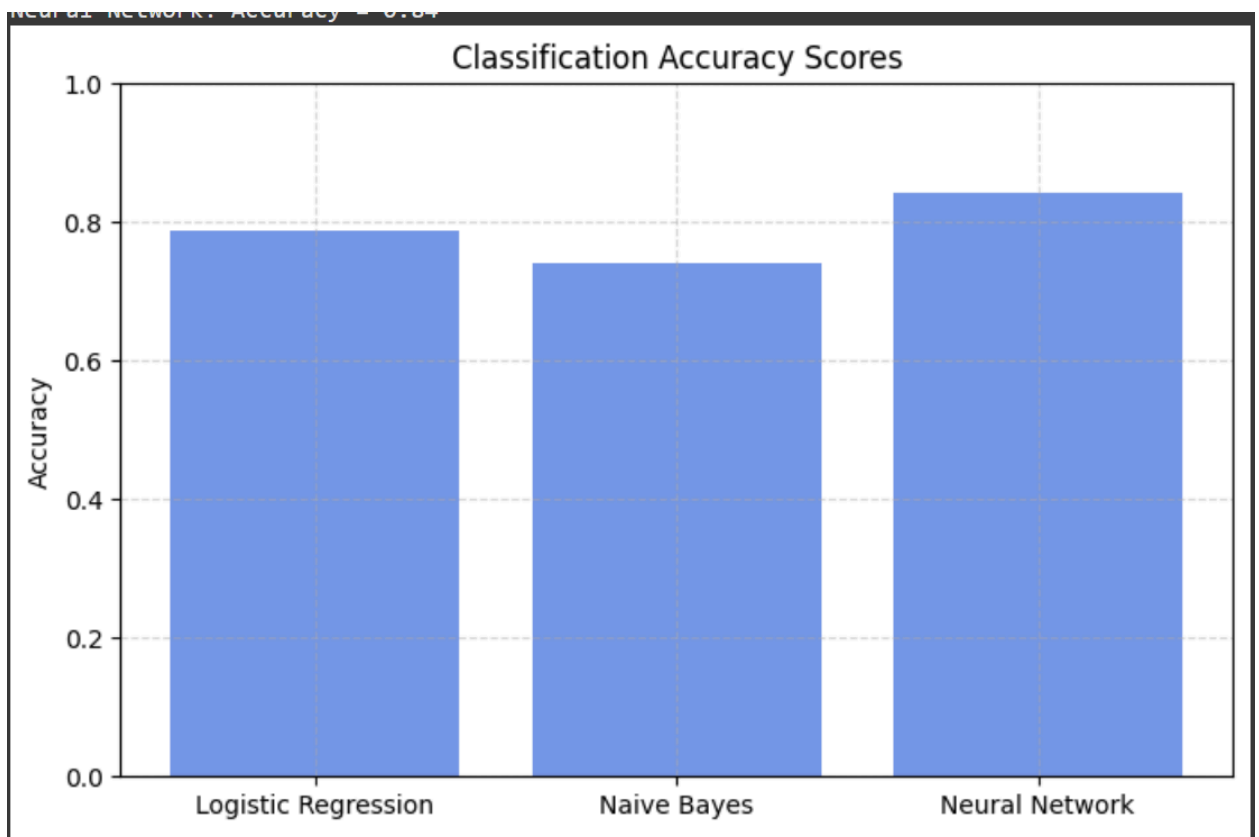
Classification Model Evaluation

To evaluate how well the classification models predicted whether a given day would result in **high** or **low food waste**, several performance metrics and visual tools were used:

1. Prediction Accuracy (Bar Chart)

A bar chart was plotted to compare the **accuracy scores** of all classification models. The models included:

- Logistic Regression
- Naive Bayes
- Neural Network (MLPClassifier)



Each model was trained on 70% of the dataset and tested on the remaining 30%. The accuracy metric helped measure how many predictions the model got correct out of all test samples.

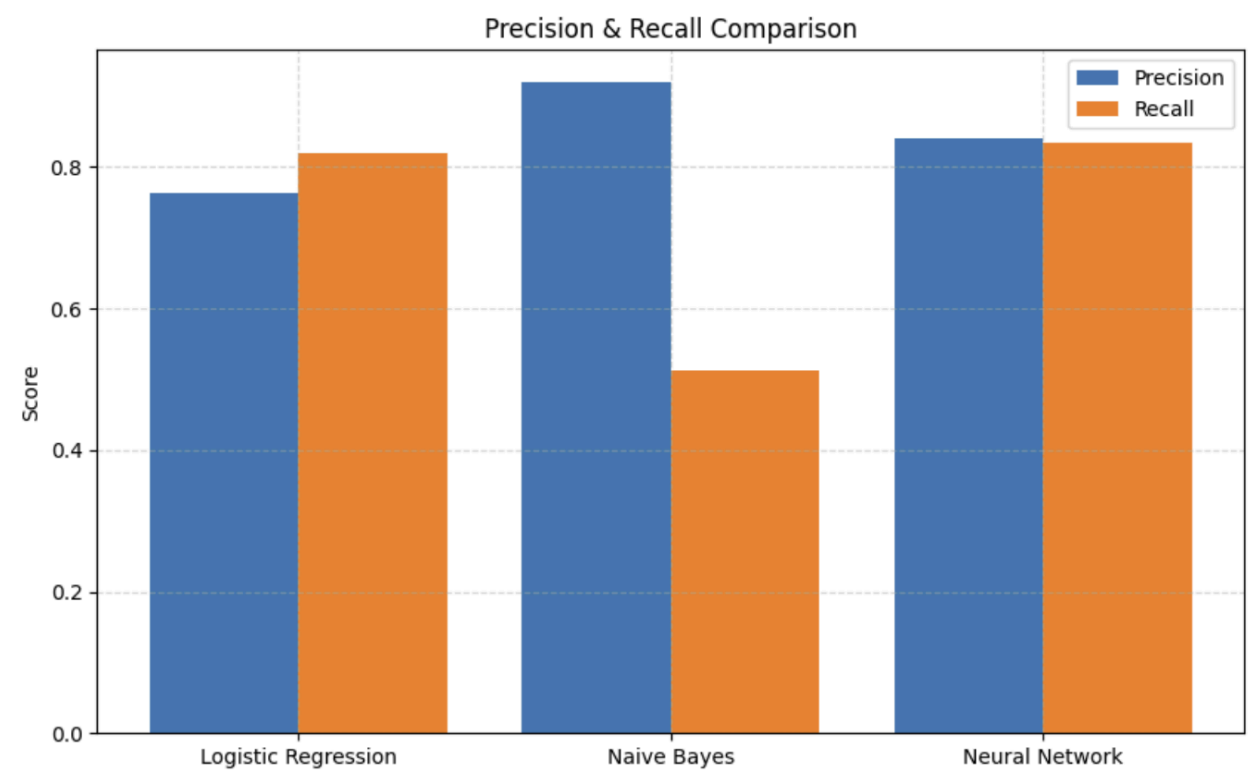
Observation:

The **Neural Network** achieved the highest accuracy, followed closely by Logistic Regression. Naive Bayes performed moderately well.

2. Precision & Recall Comparison

To further evaluate model effectiveness in identifying high-waste days, a side-by-side **bar chart of precision and recall scores** was created.

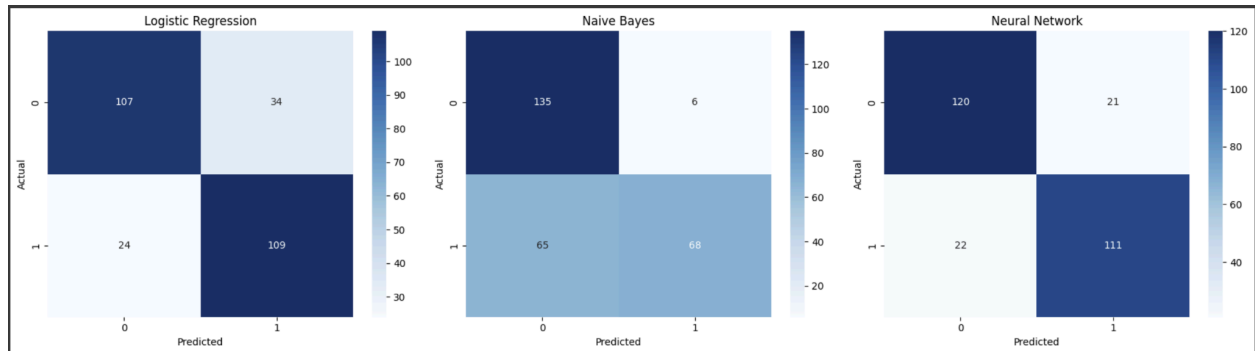
- **Precision** indicates how many of the days predicted as "high waste" were actually high.
- **Recall** shows how many actual "high waste" days were correctly identified.

**Observation:**

The Neural Network achieved the **best balance** between precision and recall, which is essential when both false positives and false negatives carry consequences (e.g., over-preparing meals or underestimating demand).

3. Confusion Matrix

For each classification model, a **confusion matrix** was generated and visualized using heatmaps. This matrix gives a detailed breakdown of true positives, true negatives, false positives, and false negatives.

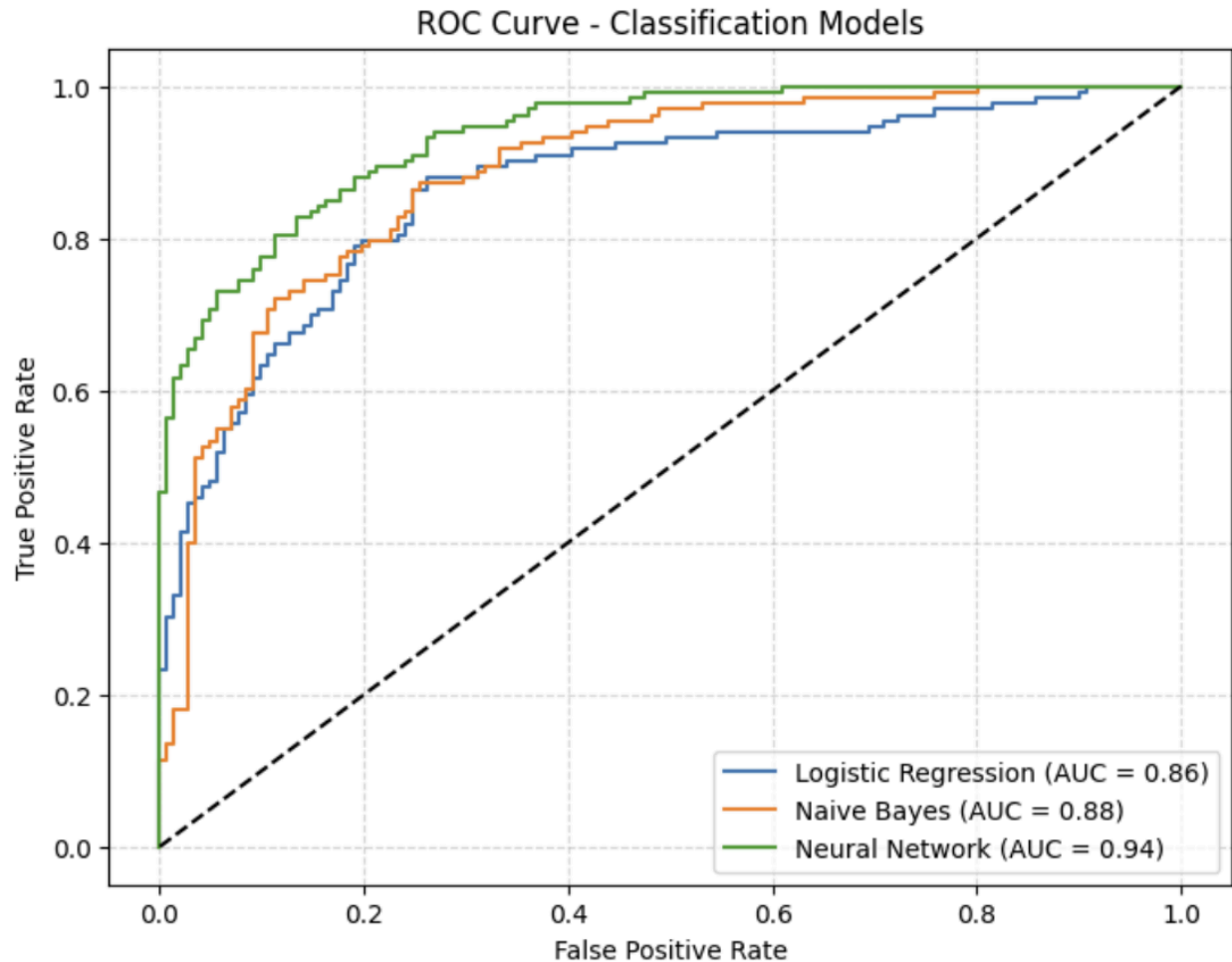


Observation:

All three models showed relatively balanced matrices, but the **Neural Network** had the **lowest number of false predictions** compared to others.

4. AUC Score & ROC Curve

The **ROC (Receiver Operating Characteristic) curve** plots the trade-off between true positive rate and false positive rate. The **Area Under the Curve (AUC)** indicates the overall classification performance.



Observation:

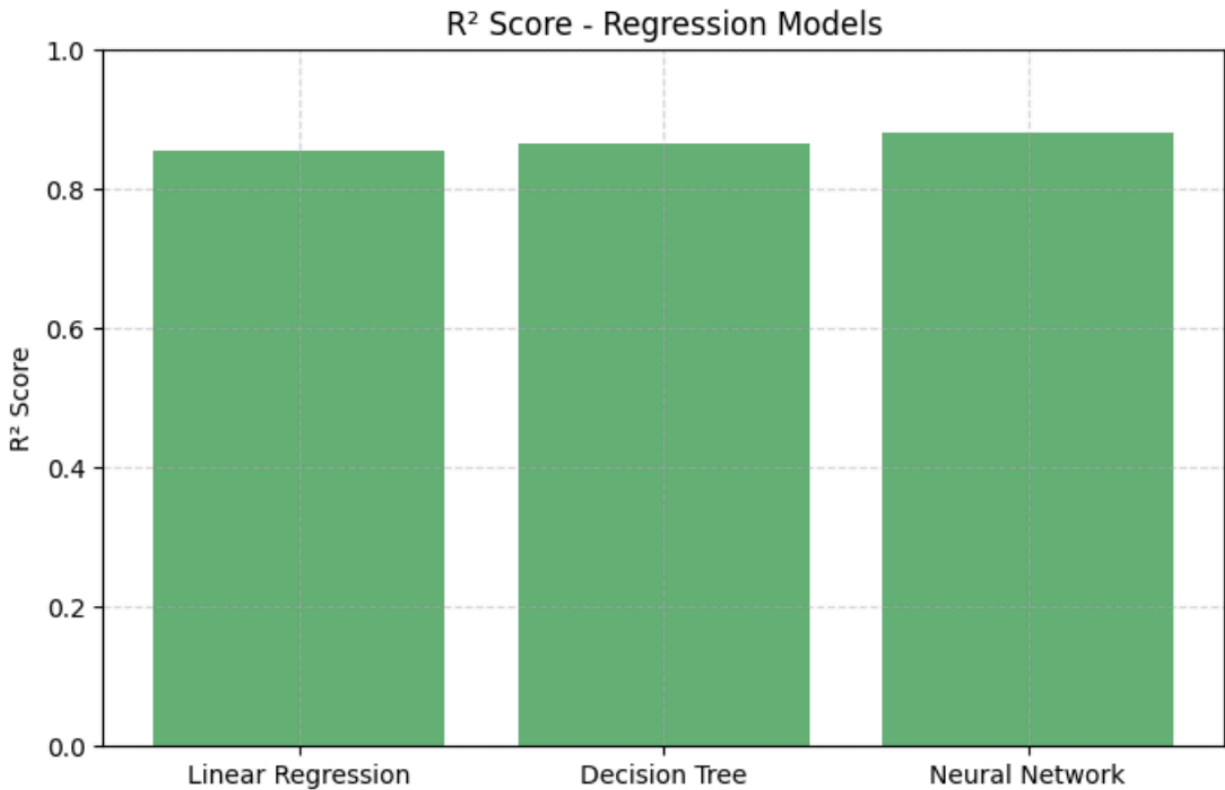
The Neural Network had the **highest AUC**, closely followed by Logistic Regression, indicating superior overall classification capability.

Regression Model Evaluation

For predicting the **exact amount of food waste (in kg)**, regression models were assessed using two primary metrics:

1. R² Score (Bar Chart)

The **R² Score** indicates how well the independent variables explain the variability in the target variable (**food_waste_kg**).



Observation:

The **Neural Network Regressor** achieved the highest R² score (≈ 0.88), showing it could explain most of the variance in the food waste data.

2. Mean Squared Error (Loss)

The **Mean Squared Error (MSE)** measures how far the model's predictions deviate from actual values. A **lower MSE** means higher prediction accuracy.

Observation:

Again, the Neural Network had the **lowest loss**, followed by Decision Tree and Linear Regression. This confirmed the network's ability to model complex, non-linear relationships in the data.



Conclusion of Comparison

- **Neural Networks** consistently outperformed other models in both regression and classification tasks.
- **Logistic Regression** and **Decision Tree** served as strong baseline models.
- The use of **multiple metrics and visualizations** ensured a reliable and interpretable model selection process.

This comparison analysis not only validates the model choice but also highlights the effectiveness of deep learning models in real-world predictive analytics

Conclusion

This project aimed to predict food waste in large-scale kitchen operations using machine learning models. By approaching the problem through both regression and classification methods, we were able to not only predict **exact food waste quantities** but also classify days into **high** or **low waste categories**, enabling more informed decision-making.

Key Insights from Results

- The **Neural Network** consistently outperformed traditional models in both regression and classification tasks. It achieved the **highest R^2 score (~0.88)** in regression and the **highest accuracy and AUC** in classification.
- **Logistic Regression** and **Decision Tree** also performed reasonably well and served as reliable baselines for comparison.
- Features such as **meals_served**, **kitchen_staff**, and **past_waste_kg** had the strongest influence on predicting food waste, as shown in the correlation analysis.

Interpretation of Model Performance

- The superior performance of the Neural Network can be attributed to its ability to **capture complex, non-linear relationships** within the data — which simpler models like Linear or Logistic Regression may not fully leverage.
- Traditional models like Linear Regression worked well for quickly interpreting feature relationships, but fell short in terms of accuracy compared to more sophisticated models.

Challenges Faced

- **Handling missing values** required careful imputation to avoid distorting the dataset, particularly for the **staff_experience** column.
- Balancing **classification labels** while ensuring model fairness was a key concern, although the dataset turned out to be relatively balanced.
- **Feature scaling** was crucial but sometimes sensitive in affecting Neural Network convergence. In some runs, especially with MLPClassifier, **convergence warnings** were encountered due to limited iterations or learning rate.

- Determining the right **architecture and hyperparameters** for the Neural Network required experimentation.

Final Comments

Overall, the project successfully demonstrated how machine learning can be applied to **reduce food waste, optimize kitchen planning**, and support **sustainable practices**. The combination of exploratory data analysis, structured preprocessing, and comprehensive model evaluation ensured that the insights drawn were both data-driven and practical.

In future work, incorporating **external factors** like customer attendance forecasts or menu type could further enhance prediction accuracy.